

TECHNOLOGY AT YOUR SERVICE

Ball-shaped camera to put 360° photography on the map

Ever since hardware entrepreneur Jonas Pfeil first started working on Panono camera, a ball-shaped device that captures photo spheres, back in 2011, the field of 360° imagery has exploded.



The Panono has a diameter of 11cm and weighs approximately 480gm

The Panono captures photos when thrown upwards, taking the shot at the moment it reaches its highest point. The 360° images it captures are technically 108MP, stitched together from the array of 3MP lenses that dot the exterior of the ball.

The camera is about the size of a grapefruit and feels just as heavy. The plastic exterior is sprinkled with small, slightly indented lenses, and it has a rubberised green trim, which provides a better grip. On one of the ball's poles there is an LED and a trigger button; on the other is a cap that covers the micro-USB port and connector for a removable handle. The handle is for situations when you do not want to throw the camera; simply attach it and click the button on the grip.

Electronic skin sensors to control mobile gadgets

A skin-worn sensor that turns the human body into a touch-sensitive surface for controlling mobile devices has been developed by scientists in Germany. Named iSkin, the sensor is made from bio-compatible silicone rubber with pressure-sensitive sensors that are stuck to the skin of the users, allowing them to use their own body to control mobile devices.



iSkin by Max Planck Institute

Developed by scientists at Max Planck Institute for Informatics and Saarland University, the experimental system has been produced in different shapes and sizes to suit various locations on the body, such as the finger, forearm or behind the ear-lobe. It is capable of detecting touch input pressure even while being stretched or bent. With the current prototypes, wearers can answer incoming calls, play music and adjust volume.

The base material is poly-dimethyl-siloxane (PDMS), an easy-to-process silicone based organic polymer. Conductive carbon black powder is added to the liquid silicone before it is spread flat by a thin-film applicator. After creating the tattoo-like designs on a computer, a laser cutter traces out the design, which makes up the sensor. It is then sandwiched between two clear sheets of silicone. The stickers are attached to the body using a medical-grade adhesive that can be easily peeled off after use, without hurting the skin.

Insect-sized robot can jump on water

Researchers from Harvard University have built an insect-sized robot that mimics the way water striders jump on water. By observing water striders using high-speed cameras, scientists noticed that insects do not simply push down on the water but gradually accelerate their legs so as to not break surface tension. Striders also sweep their legs inwards before each jump, to maximise the amount of time they touch the surface, which increases the force of their pushes.

The researchers used these principles to develop an ultra-lightweight robot with a 2cm-long body inspired by origami. Its 5cm-long wire legs are curved at the tips like a real water strider's and coated with a material that repels water.

A flea-inspired jumping system, called a torque reversal catapult, launches the robot from the surface of the water up to 14.2cm in the air.